

Fatih Sultan Mehmet Vakıf Üniversitesi

Computer Network Concepts

PROJECT REPORT

Backgammon Multiplayer Game

Project No: 2

Prepared by:

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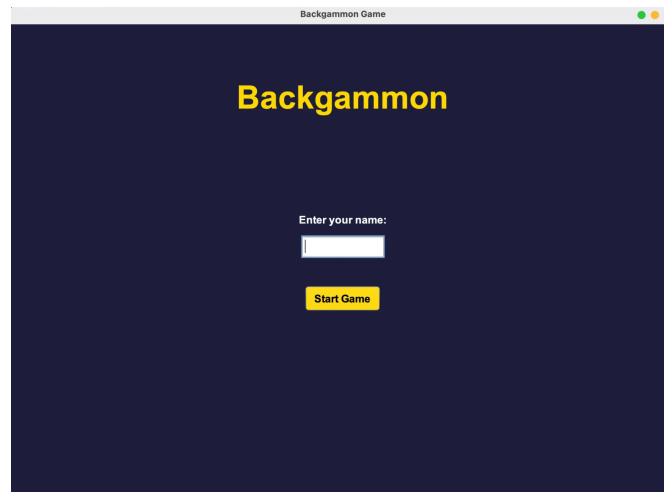
2026

Instructor: Dr. Samet Kaya
Research Assistant: Ertuğrul İslamoğlu

1. Introduction

This report documents the design and implementation of a multiplayer Backgammon game developed as part of the Computer Network Concepts course project. The project demonstrates the practical application of client-server socket programming using Java.

The game allows two players to connect over a network and play Backgammon in real-time. The server is hosted on Amazon Web Services (AWS) and manages communication between the two clients.



Start Screen of the Application

2. System Architecture

The system follows a Client-Server architecture consisting of three main components:

2.1 GameServer

The server acts as a relay between the two players. It listens on port 6000 for incoming connections. Once two players connect, it forwards each player's moves to the other player in real-time using threads.

- Hosted on AWS EC2 instance (IP: 56.228.22.133)
- Runs on port 6000
- Supports multiple simultaneous games — each pair of players is handled in a separate thread
- Uses two threads per game session to handle both players simultaneously
- Assigns player numbers (Player 1 / Player 2) upon connection

2.2 GameClient

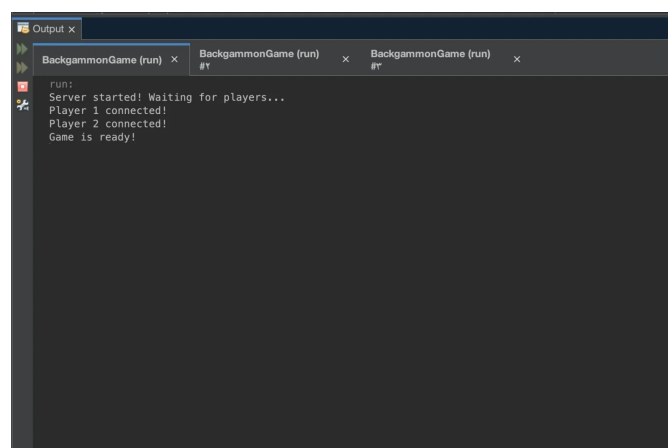
Each player runs the client application on their machine. The client connects to the server using the server's IP address and port 6000. It sends moves to the server and receives the opponent's moves in a background thread.

2.3 Communication Protocol

Moves are transmitted as plain text strings in the following format:

```
MOVE:fromIndex:toIndex
```

For example, MOVE:12:15 means a piece was moved from point 13 to point 16.



Server Console Output — Players Connected

3. Implementation

3.1 Game Logic (BackgammonBoard)

The game logic is encapsulated in the BackgammonBoard class. The board is represented as an integer array of 24 elements:

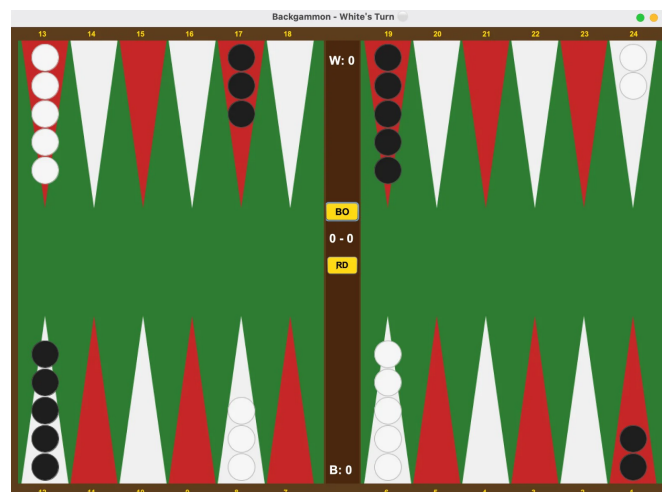
- Positive values represent Player 1 (white) pieces
- Negative values represent Player 2 (black) pieces
- Zero means the point is empty

Key methods include: isValidMove(), movePiece(), rollDice(), canBearOff(), hasWon(), and enterFromBar().

3.2 GUI Design

The graphical interface was built using Java Swing. An innovative approach was used for rendering the board: instead of drawing shapes dynamically, 48 pre-generated PNG images were created using Java's Graphics2D API — one image for each combination of triangle color (red/white), piece color (white/black), piece count (0–5), and orientation (normal/flipped).

Each of the 24 triangle points is represented as a JLabel displaying the corresponding image. When a move is made, only the image source is updated — no actual dragging occurs.

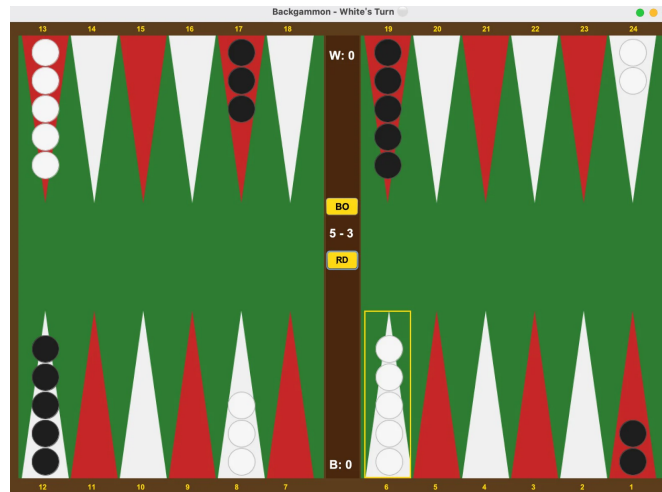


Game Board — Initial Setup with Pieces

3.3 Game Rules Implemented

- Dice roll — random values between 1 and 6
- Move validation — direction, distance, and blocking rules
- Hitting (eating) opponent's blot — piece sent to the Bar
- Bar re-entry — player must re-enter before any other move
- Bearing off — when all pieces are in the home board
- Win detection — when all 15 pieces are borne off

- Turn-based system — dice can only be rolled once per turn
- Two-dice movement — each die value can be used for a separate move



Gameplay Screenshot — Piece Being Moved

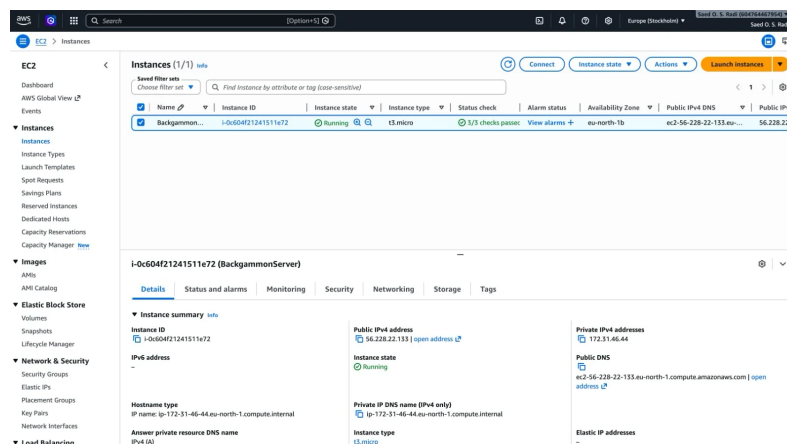
4. AWS Deployment

The GameServer was deployed on an Amazon EC2 instance running Amazon Linux 2023. The following steps were performed:

- Created an EC2 t3.micro instance (Free Tier eligible)
- Installed Java 21 (Amazon Corretto) on the instance
- Opened port 6000 in the Security Group inbound rules
- Uploaded and compiled GameServer.java via SCP
- Started the server in the background using nohup

Server IP Address: 56.228.22.133

The client applications connect to this IP address directly, satisfying the project requirement for AWS deployment.



AWS EC2 Console — Instance Running

5. Version Control (GitHub)

The project was managed using Git and hosted on a private GitHub repository. Commits were made regularly throughout the development process to track progress.

Repository: github.com/SaadOsama10/backgammon-network-project

Commits		
History for backgammon-network-project / BackgammonGame on <code>main</code>		
All users All time		
Commits on May 6, 2026		
Update server to support multiple simultaneous games	11902ac	
Commits on May 5, 2026		
Fix opponent move display and server connection	05ae006	
Commits on May 2, 2026		
Add code comments to all Java files	0b92767	
Add AWS server deployment	b402764	
Commits on Apr 17, 2026		
Fix bar entry validation and dice roll restriction	adeb4c2	
Commits on Apr 16, 2026		
dd multiplayer client-server connection	330f9ac	
Add win condition and end screen	4c22e1e	
Add bearing off and bar display	a7d6f9b	
Add move direction validation and bar logic	d778d4b	

GitHub Repository — Commits History

6. Conclusion

This project successfully demonstrates the implementation of a fully functional multiplayer Backgammon game using Java socket programming. All core requirements were met:

- Server hosted on AWS with real IP address
- Two clients communicate through the server
- Complete game rules implemented
- User-friendly graphical interface
- Start screen and end screen with replay option
- Project tracked on GitHub from the beginning

The project provided valuable hands-on experience with network programming concepts including socket communication, multi-threading, and client-server architecture.